



AICT

Inside the System Unit: Components & Computing

Presented By:

- **Abdullah Waqar**
- **Ahmad Raza**
- **Naeem Ilyas**
- **Daniyal Shahid**

Presented to: Mr Zaheer Hussain Fida



Inside the Case & Motherboard

Presenter: Abdullah Waqar

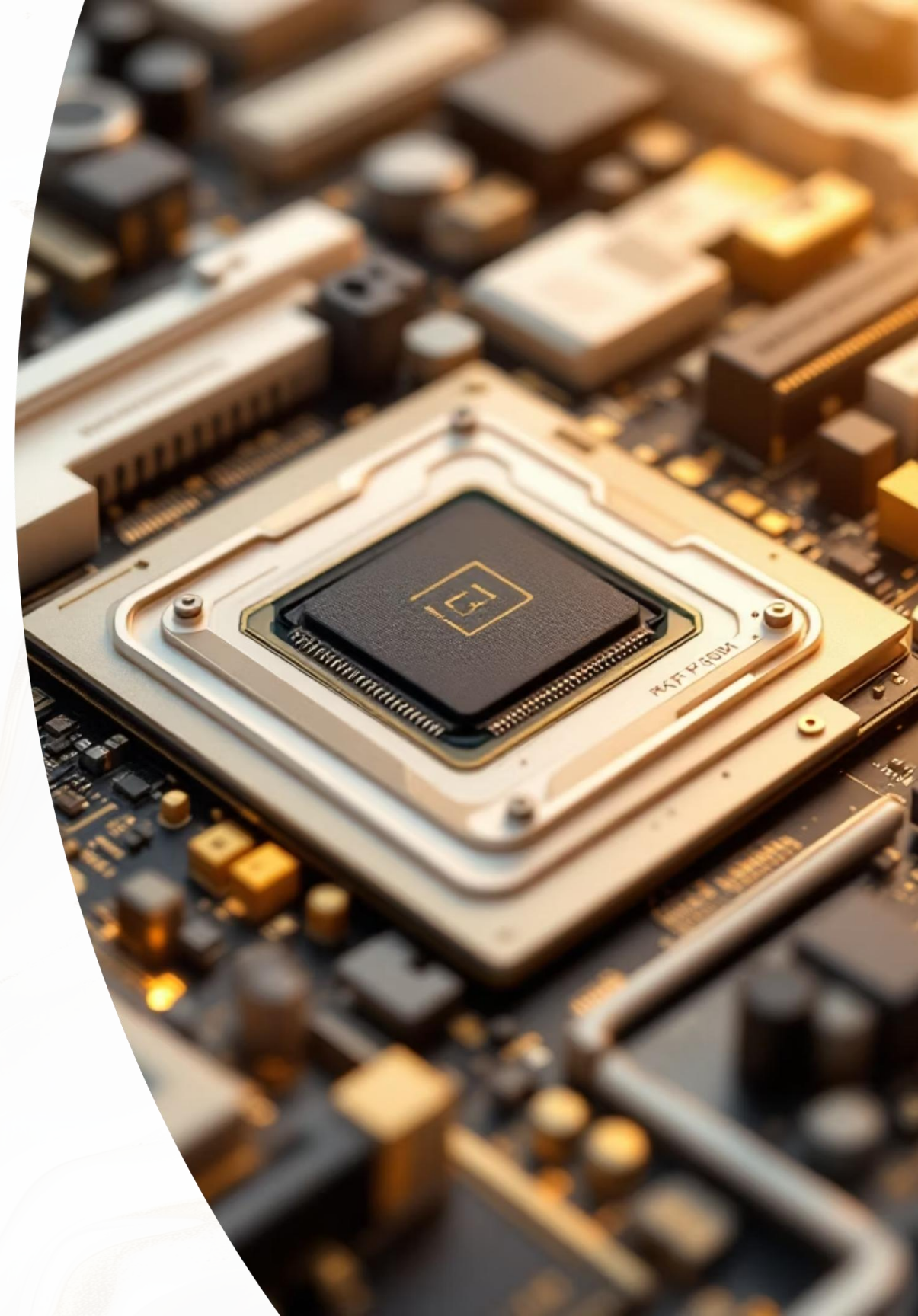
Inside the Case

The system unit is the main container that holds all internal components working together.

The Motherboard

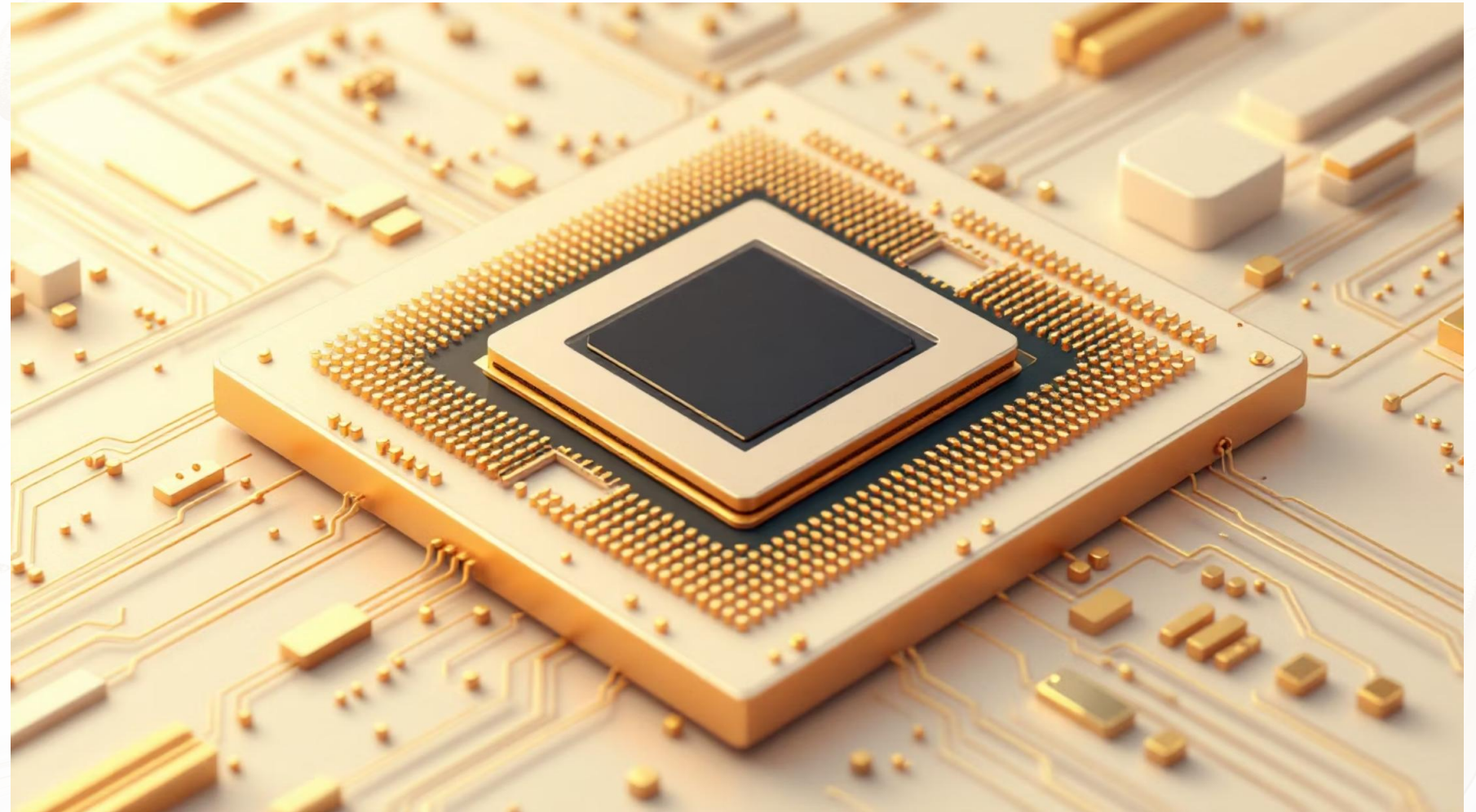
The main circuit board of the computer.

- **It connects all parts (CPU, RAM, Hard Drive) and allows them to communicate.**
- **Without it, a computer cannot function.**



The Processor (CPU)

Presenter: Abdullah Waqar



The Brain

Known as the Central Processing Unit (CPU).

Main Functions:

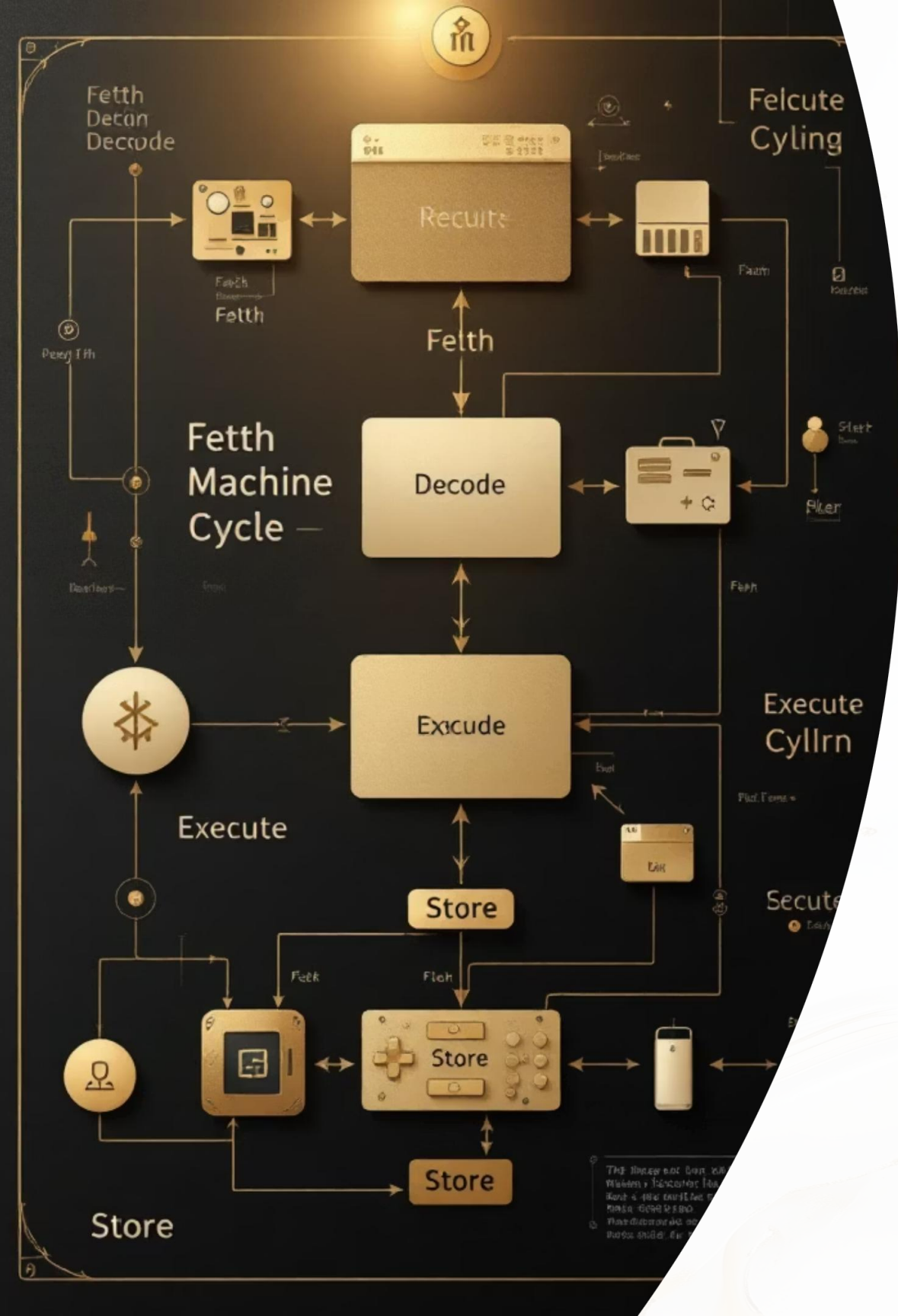
Fetch: Getting instructions from memory.

Decode: Translating instructions into commands.

Execute: Carrying out the commands.

Key Components

ALU (Arithmetic Logic Unit) and Control Unit.



Control Unit & ALU (Machine Cycle)

Presenter: Ahmad Raza

Control Unit (CU)

Directs the flow of data and manages the operation of the processor.

ALU

Handles mathematical calculations and logical comparisons.

The Machine Cycle

The continuous process of fetching, decoding, executing, and storing data. It's how the CPU processes every single task.



Registers, System Clock & Mobile Devices

Presenter: Naeem Ilyas



Registers

High-speed storage locations inside the CPU that hold data temporarily for immediate processing.



System Clock

Controls the timing of all computer operations (measured in GHz).



Computing Devices

The same concepts apply to Personal Computers and Mobile Devices, just in a more compact, power-efficient form.



Cooling, IoT & Cloud Computing

Presenter: Daniyal



Processor Cooling

Uses heat sinks and fans to prevent the CPU from overheating.



Internet of Things (IoT)

A network of physical objects (smart bulbs, watches) connected to the internet.



Cloud Computing & Services

Accessing computing services (servers, storage, software) over the internet instead of local hardware.

***Services:* Google Drive, iCloud, AWS.**

